

Network+ N10-009 Memorization Sheet

Print this and keep it beside you while you study. Every number, order, and pairing on this sheet is fair game on the exam. Nothing here needs understanding to score; it needs recall.

01 The exam itself

FACT	VALUE
Exam code	N10-009 (Network+)
Questions	Up to 90, multiple choice plus performance-based
Time	90 minutes
Passing score	720 on a scale of 100 to 900 (about 72%)
1.0 Networking Concepts	23%
2.0 Network Implementation	20%
3.0 Network Operations	19%
4.0 Network Security	14%
5.0 Network Troubleshooting	24%

Read for the qualifier: MOST likely, BEST, FIRST, NEXT. Eliminate two options, then choose. Never leave a question blank.

02 Ports and protocols

SERVICE	PORT	TRANSPORT	NOTES AND SECURE FORM
FTP	20 data, 21 control	TCP	Plaintext. Secure: SFTP on 22, or FTPS on 989/990
SSH, SFTP, SCP	22	TCP	Encrypted shell and file copy. Replaces Telnet and FTP
Telnet	23	TCP	Plaintext. Replace with SSH
SMTP	25	TCP	Server to server mail. Secure submission: 587
TACACS+	49	TCP	Device administration AAA , encrypts the whole payload
DNS	53	UDP queries, TCP zone transfers	Secure: DoT 853, DoH 443
DHCP	67 server, 68 client	UDP	DORA . DHCPv6 uses 546 client, 547 server
TFTP	69	UDP	No authentication. Firmware and config transfers
HTTP	80	TCP	Secure: HTTPS 443
Kerberos	88	TCP or UDP	Windows domain authentication. Breaks if clocks differ by more than 5 minutes
POP3	110	TCP	Secure: POP3S 995
NTP	123	UDP	Time sync. Secured with NTS
IMAP	143	TCP	Secure: IMAPS 993
SNMP	161 agent (polls), 162 manager (traps)	UDP	v1 and v2c use a community string; v3 adds authentication and encryption

SERVICE	PORT	TRANSPORT	NOTES AND SECURE FORM
LDAP	389	TCP	Secure: LDAPS 636
HTTPS	443	TCP	TLS web traffic; also carries DoH
SMB	445	TCP	Windows file and printer sharing
IKE (IPsec)	500, NAT-T 4500	UDP	Key exchange. ESP is IP protocol 50, AH is 51
Syslog	514	UDP	Central logging. Severity 0 to 7
SMTPS, submission	587	TCP	Mail client to server with TLS
LDAPS	636	TCP	LDAP over TLS
DNS over TLS	853	TCP	Encrypted resolver queries
IMAPS	993	TCP	IMAP over TLS
POP3S	995	TCP	POP3 over TLS
SQL Server	1433	TCP	Microsoft SQL
RADIUS	1812 auth, 1813 accounting	UDP	Network access AAA (Wi-Fi , VPN , 802.1X). Legacy 1645/1646
MySQL	3306	TCP	MySQL and MariaDB
RDP	3389	TCP	Remote Desktop
SIP	5060 plain, 5061 TLS	TCP or UDP	Sets up voice and video calls

Encrypted pairs

- **HTTP** 80 to **HTTPS** 443
- Telnet 23 to **SSH** 22
- **FTP** 21 to **SFTP** 22
- **LDAP** 389 to **LDAPS** 636
- **IMAP** 143 to **IMAPS** 993
- **POP3** 110 to **POP3S** 995
- **SMTP** 25 to **SMTPS** 587
- **DNS** 53 to **DoT** 853 or **DoH** 443
- **SNMP** v1/v2c to **SNMPv3** (same ports)
- **SIP** 5060 to **SIP** over **TLS** 5061

Port ranges and IP protocol numbers

- 0 to 1023: well-known
- 1024 to 49151: registered
- 49152 to 65535: dynamic (client) ports
- **ICMP** 1, **TCP** 6, **UDP** 17
- **GRE** 47, **ESP** 50, **AH** 51
- **OSPF** 89, **EIGRP** 88 (ride directly on IP)

03 OSI and TCP/IP models

LAYER	PDU	KEY IDEA	DEVICES AND PROTOCOLS
7 Application	Data	Protocols users touch	HTTP , HTTPS , SMTP , DNS , DHCP , FTP , SSH , SNMP , Telnet; proxies, web application firewalls
6 Presentation	Data	Formatting, encryption, compression	TLS , JPEG , MPEG , ASCII , Unicode
5 Session	Data	Start, maintain, end dialogs	NetBIOS, RPC , session control
4 Transport	Segment (TCP), datagram (UDP)	Port numbers; reliable vs fast	TCP , UDP ; port-filtering firewalls, load balancers
3 Network	Packet	IP addresses, TTL , routing	Routers, Layer 3 switches; IP , ICMP , IPsec , OSPF , RIP

LAYER	PDU	KEY IDEA	DEVICES AND PROTOCOLS
2 Data Link	Frame	MAC addresses, MAC and LLC sublayers, frame check sequence	Switches, NICs, access points, bridges; Ethernet, 802.11, ARP, STP, 802.1Q
1 Physical	Bit	Signals, cables, connectors, radio	Hubs, repeaters, transceivers, media converters, cables

- Bottom up: Please Do Not Throw Sausage Pizza Away
- Top down: All People Seem To Need Data Processing
- TCP/IP model: Application (OSI 5 to 7), Transport (4), Internet (3), Network Access or Link (1 and 2)
- Encapsulation: data, segment, packet, frame, bits. Capture order: Ethernet header (MACs), IP header (addresses, TTL), TCP or UDP header (ports), payload
- MAC address: 48 bits, 12 hex digits, first 24 bits are the OUI. Rewritten at every router hop; the IP address stays the same end to end
- MTU 1500 bytes standard; jumbo frames about 9000
- TCP handshake: SYN, SYN-ACK, ACK. Close: FIN and ACK. Abort: RST
- Traffic types: unicast one to one; broadcast one to all on the segment (routers stop it; IPv6 has none); multicast one to a group (224.0.0.0 to 239.255.255.255); anycast one address on many servers, nearest answers

04 IPv4 addressing and subnetting

CLASS	FIRST OCTET	DEFAULT MASK	USE
A	1 to 126	/8, 255.0.0.0	Huge networks
B	128 to 191	/16, 255.255.0.0	Medium networks
C	192 to 223	/24, 255.255.255.0	Small networks
D	224 to 239	none	Multicast (OSPF uses 224.0.0.5 and 224.0.0.6)
E	240 to 255	none	Experimental
127	127.0.0.0/8	loopback	127.0.0.1 is this host

RANGE	MEANING
10.0.0.0/8 (10.0.0.0 to 10.255.255.255)	Private, RFC 1918
172.16.0.0/12 (172.16.0.0 to 172.31.255.255)	Private, RFC 1918. 172.32.x.x is public
192.168.0.0/16 (192.168.0.0 to 192.168.255.255)	Private, RFC 1918
169.254.0.0/16	APIPA, link-local. Means DHCP failed
255.255.255.255	Limited broadcast (DHCP Discover)
0.0.0.0/0	Default route, route of last resort
100.64.0.0/10	Carrier-grade NAT shared space

PREFIX	MASK	BLOCK SIZE	USABLE HOSTS
/8	255.0.0.0	1 in octet 1	16,777,214
/16	255.255.0.0	1 in octet 2	65,534
/17	255.255.128.0	128 in octet 3	32,766
/18	255.255.192.0	64 in octet 3	16,382
/19	255.255.224.0	32 in octet 3	8,190
/20	255.255.240.0	16 in octet 3	4,094

PREFIX	MASK	BLOCK SIZE	USABLE HOSTS
/21	255.255.248.0	8 in octet 3	2,046
/22	255.255.252.0	4 in octet 3	1,022
/23	255.255.254.0	2 in octet 3	510
/24	255.255.255.0	1 in octet 3	254
/25	255.255.255.128	128	126
/26	255.255.255.192	64	62
/27	255.255.255.224	32	30
/28	255.255.255.240	16	14
/29	255.255.255.248	8	6
/30	255.255.255.252	4	2 (point-to-point links)
/31	255.255.255.254	2	2, no broadcast (point-to-point only)
/32	255.255.255.255	1	A single host

- Mask octet values: 128, 192, 224, 240, 248, 252, 254, 255 = 1 to 8 network bits
- Binary place values: 128, 64, 32, 16, 8, 4, 2, 1
- Powers of two: 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, 2048, 4096
- Usable hosts = 2 to the host bits, minus 2. Subnets created = 2 to the borrowed bits
- Hosts needed: add 2, round up to the next power of two, that many host bits
- A subnet must start on a multiple of its own block size (192.168.10.96/26 is invalid; 64 and 128 are valid starts)
- Summarization: 10.1.0.0/24 through 10.1.3.0/24 = 10.1.0.0/22

Block size method

- 1. Find the interesting octet: the first mask octet that is not 255 or 0.
- 2. Block size = 256 minus that mask octet (.224 gives 32, .240 gives 16, .192 gives 64).
- 3. Subnets start at multiples of the block size: 0, 32, 64, 96, 128, 160, 192, 224 for a block of 32.
- 4. Network = the multiple at or below the host. Broadcast = one less than the next multiple. Usable = everything between.
- Example: 192.168.1.100/27. Block 32. Network 192.168.1.96, broadcast .127, hosts .97 to .126.

05 IPv6

PREFIX	TYPE	NOTES
2000::/3	Global unicast	Routable on the internet (addresses starting 2 or 3)
fe80::/10	Link-local	Automatic on every interface, never routed. Used by NDP and routing protocols
fc00::/7 (in practice fd..)	Unique local	Private addressing, routable inside the organization only
ff00::/8	Multicast	ff02::1 all nodes, ff02::2 all routers. No broadcast in IPv6
::1	Loopback	This host
::	Unspecified	No address yet
2001:db8::/32	Documentation	Used in examples only

- 128 bits, eight groups of four hex digits. A standard **LAN** is a /64: 64 bits network, 64 bits interface **ID**
- Shortening: drop leading zeros in each group; replace one run of all-zero groups with :: (only once)
- SLAAC**: the router advertisement supplies the /64 prefix and the host builds its own interface **ID**
- EUI-64**: split the 48-bit **MAC**, insert ff:fe in the middle, flip the seventh bit

- DHCPv6: stateful hands out addresses; stateless hands out only options such as **DNS**
- **NDP** replaces **ARP**: neighbor solicitation and neighbor advertisement over ICMPv6 multicast
- Dual stack runs both. Tunneling (6to4, **ISATAP**, Teredo, **GRE**) carries v6 inside v4. **NAT64** with **DNS64** translates for v6-only clients

06 Cabling, connectors, and Ethernet standards

COPPER	SPEED	DISTANCE	NOTES
Cat 5e	1 Gbps	100 m	Minimum for gigabit
Cat 6	1 Gbps; 10 Gbps	100 m; 55 m at 10 Gbps	
Cat 6A	10 Gbps	100 m	The answer for 10 Gbps at full distance
Cat 7	10 Gbps	100 m	Shielded, non-RJ45 connectors (GG45 , TERA), rare
Cat 8	25 or 40 Gbps	30 m	Short data center runs
Coax RG-6	Cable modem, video		F-type connector; BNC on legacy LANs
Twinax (DAC)	10 to 100 Gbps	A few meters	Transceivers permanently attached; cheaper than optics

- All twisted pair Ethernet: 100 meters (328 feet) including patch cords
- **RJ45** is **8P8C** for Ethernet; **RJ11** for telephone
- **UTP** default; **STP** for high-EMI areas (motors, fluorescent ballasts)
- **T568B** pins: 1 white-orange, 2 orange, 3 white-green, 4 blue, 5 white-blue, 6 green, 7 white-brown, 8 brown. **T568A** swaps the orange and green pairs
- Same standard both ends = straight-through (host to switch). Different = crossover (switch to switch without auto-**MDIX**)
- Plenum-rated for air-handling spaces above drop ceilings; riser-rated for vertical runs between floors
- **CSMA/CD**: listen, send, back off on collision. Full duplex has no collisions. Duplex mismatch = late collisions and **CRC** errors

FIBER	CORE	SOURCE	REACH	JACKET
Single-mode (SMF)	about 9 microns	Laser	10 km and beyond	Yellow
Multimode (MMF)	50 or 62.5 microns	LED or VCSEL	About 300 to 550 m at 10 Gbps	Orange (OM1 , OM2), aqua (OM3 , OM4)

- **LC**: small square snap-in, most common on transceivers. **SC**: larger square push-pull. **ST**: round bayonet twist, legacy. **MPO/MTP**: 12 or 24 fibers in one plug for 40 and 100 Gbps
- **UPC** blue flat polish; **APC** green angled polish (less reflection, long haul and **PON**). Never mate **UPC** to **APC**
- Media converter changes medium without changing data (fiber to copper, **SMF** to **MMF**)
- Transceivers: **SFP 1 Gbps**, **SFP+ 10 Gbps**, **QSFP/QSFP+ 40 Gbps**, **QSFP28 100 Gbps**, **QSFP-DD** and **OSFP 400 Gbps**. Both ends must match speed, wavelength, and fiber type

STANDARD	SPEED	MEDIUM	DISTANCE
10BASE-T	10 Mbps	Cat 3 or better	100 m
100BASE-TX	100 Mbps	Cat 5 or better	100 m
1000BASE-T	1 Gbps	Cat 5e or better	100 m
10GBASE-T	10 Gbps	Cat 6A (Cat 6 to 55 m)	100 m
40GBASE-T	40 Gbps	Cat 8	30 m
1000BASE-SX	1 Gbps	Multimode	About 550 m
1000BASE-LX	1 Gbps	Single-mode (or MMF with mode conditioning)	10 km
10GBASE-SR	10 Gbps	Multimode	About 300 m (400 m on OM4)
10GBASE-LR	10 Gbps	Single-mode	10 km

STANDARD	SPEED	MEDIUM	DISTANCE
10GBASE-ER	10 Gbps	Single-mode	40 km

Naming: S = short reach, multimode. L = long reach, single-mode. E = extended reach. T = twisted pair. **SR** on single-mode fiber, or **SR** at 2 km, is a mismatch.

07 Wireless

STANDARD	NAME	BAND	MAX RATE	KEY FEATURE
802.11a		5 GHz	54 Mbps	OFDM, short range
802.11b		2.4 GHz	11 Mbps	Long range, slow
802.11g		2.4 GHz	54 Mbps	Backward compatible with b
802.11n	Wi-Fi 4	2.4 and 5 GHz	600 Mbps	MIMO, 40 MHz channels
802.11ac	Wi-Fi 5	5 GHz only	Gigabit class (up to 6.9 Gbps)	MU-MIMO, 80 and 160 MHz channels
802.11ax	Wi-Fi 6 / 6E	2.4, 5, and 6 GHz (6E)	About 9.6 Gbps	OFDMA for crowded spaces
802.11be	Wi-Fi 7	2.4, 5, and 6 GHz	Tens of Gbps	320 MHz channels, multi-link operation

- 2.4 GHz: non-overlapping channels 1, 6, 11 (20 MHz). Longest range, most interference (microwaves, Bluetooth, neighbors)
- 5 GHz: many channels, 40/80/160 MHz bonding, DFS channels must yield to radar. 6 GHz: Wi-Fi 6E only, shortest range
- Wider channels = more speed, more overlap and noise. Use 20 MHz in dense 2.4 GHz deployments
- SSID = network name. BSSID = MAC of one radio on one AP. ESSID = same SSID across many APs for roaming
- Infrastructure (through an AP), ad hoc (IBSS, no AP), mesh (APs link wirelessly), point-to-point (directional antennas between buildings)
- Omnidirectional for general coverage; directional (Yagi, patch, parabolic) for long links and hallways
- Autonomous APs configured individually; lightweight APs managed by a wireless LAN controller
- Site survey produces a heat map. Overlap cells 15 to 20 percent for roaming. Band steering pushes clients to 5 GHz

SECURITY	STATUS	DETAIL
WEP	Broken	Never use
WPA	Deprecated	TKIP
WPA2	Acceptable	AES with CCMP. Personal (PSK) or Enterprise (802.1X with RADIUS)
WPA3	Current	Personal uses SAE (no offline cracking). Enterprise offers 192-bit mode
Enterprise		802.1X authentication against RADIUS, per-user credentials or certificates, per-user VLAN
Captive portal		Intercepts first web request for terms or login; guest SSID on an isolated VLAN
Hidden SSID, MAC filtering	Not security	Trivially bypassed; hardening at best

08 Switching

- Switch jobs: learn source MACs into the MAC (CAM) table, forward known destinations, flood unknowns and broadcasts, block loops with STP
- Each switch port is one collision domain. Each VLAN is one broadcast domain. A hub is one collision domain. A router separates broadcast domains
- Unmanaged: no config. Managed: VLANs, STP, port security. Layer 3 switch: also routes between VLANs in hardware

- In-band management over the network (**SSH**, web). Out-of-band via console port or dedicated management network
- Access port: one **VLAN**, untagged. Trunk port: many **VLANs** tagged with **802.1Q** (4-byte tag, **IDs** 1 to 4094)
- Native **VLAN** crosses the trunk untagged (default **VLAN** 1) and must match on both ends
- Voice **VLAN** lets a phone tag while the **PC** behind it stays untagged. Management **VLAN** holds switch interfaces
- Routing between **VLANs**: **SVI** on a Layer 3 switch, or router on a stick with one subinterface per **VLAN**
- **LACP** (**802.3ad**, **802.1AX**) bundles links; every member needs matching speed, duplex, and **VLANs**. One flow still uses one link
- **MTU** 1500; jumbo about 9000; every device in the path must match. Symptom of mismatch: small pings work, large transfers fail
- Port mirroring (**SPAN**) copies traffic to an analyzer port. Port security limits **MACs**; violation puts the port in error-disabled
- Duplex mismatch: one side hard-coded, the other auto falls to half duplex. Late collisions, **CRC** errors, slow link

SPANNING TREE	DETAIL
Standard	STP 802.1D; RSTP 802.1w; MSTP 802.1s
Root bridge	Lowest bridge ID (priority then MAC). Set core priority low
Port roles	Root port (toward root), designated port (one per link), blocked (redundant). RSTP adds alternate and backup
STP states	Blocking, listening, learning, forwarding: 30 to 50 seconds
RSTP states	Discarding, learning, forwarding: converges in seconds
PortFast	Skips listening and learning on host ports. Never on switch-to-switch links
BPDU guard	Shuts a PortFast port that receives a BPDU (rogue switch)
Root guard	Stops a downstream switch from becoming root
Loop guard	Protects against unidirectional link failures
Loop symptoms	Broadcast storm, CPU at 100 percent, MAC addresses flapping, everything slow after adding a redundant cable

POE STANDARD	NAME	AT THE PORT	DELIVERED TO DEVICE
802.3af	PoE	15.4 W	About 13 W
802.3at	PoE+	30 W	About 25.5 W
802.3bt Type 3	PoE++	60 W	About 51 W
802.3bt Type 4	PoE++	90 W	About 71 W

09 Routing, NAT, and gateway redundancy

Route selection order

- 1. Longest prefix match wins (a /24 beats a /16 beats 0.0.0.0/0).
- 2. Same prefix: lowest administrative distance wins.
- 3. Same source: lowest metric wins.

ROUTE SOURCE	ADMINISTRATIVE DISTANCE
Directly connected	0
Static	1
External BGP	20
EIGRP	90
OSPF	110
IS-IS	115

ROUTE SOURCE		ADMINISTRATIVE DISTANCE	
RIP		120	
Internal BGP		200	
PROTOCOL	TYPE	METRIC	NOTES
RIP	Distance-vector, IGP	Hop count, max 15	Updates every 30 s; RIPv2 supports CIDR ; RIPng for IPv6 ; small networks
OSPF	Link-state, IGP	Cost from bandwidth	Open standard, areas, Dijkstra SPF , fast convergence, multicast 224.0.0.5 and .6
EIGRP	Advanced distance-vector (hybrid), IGP	Bandwidth and delay	Cisco origin, fast convergence
BGP	Path-vector, EGP	AS path and policy	Between autonomous systems, runs on TCP 179 , the internet's protocol, slow by design

- Distance-vector knows what neighbors say; link-state knows the whole map (more **CPU**, faster, fewer loops)
- Static: predictable, no overhead, no adaptation. Dynamic: adapts to failures. Default route 0.0.0.0/0 is the way out
- Routers rewrap the frame (new **MACs**), keep the packet (same **IPs**), decrement **TTL** by one. **TTL 0** = discard
- Static **NAT**: one to one, publish a server. Dynamic **NAT**: pool, many to many. **PAT** (overload): many to one by source port, every home router
- Port forwarding: public port to an inside host. Inside local = private address; inside global = its public translation
- FHRP**: routers share a virtual **IP** and virtual **MAC**. **HSRP** Cisco (active/standby). **VRRP** open standard (master/backup). **GLBP** Cisco with load balancing
- Subinterfaces: one physical interface split per **VLAN** (router on a stick)
- Convergence: time for all routers to agree after a change

10 DHCP, DNS, and time

DHCP

- DORA**: Discover (broadcast from 0.0.0.0 to 255.255.255.255), Offer, Request, Acknowledge. **UDP** 67 server, 68 client
- Scope = the pool. Exclusion = addresses never leased. Reservation = fixed address pinned to a **MAC**. Lease = how long
- Clients renew at 50 percent of the lease (T1) and rebind at 87.5 percent (T2)
- Options: 3 default gateway, 6 **DNS** servers, 15 domain name, 42 **NTP**, 66 **TFTP** server name, 150 Cisco phone **TFTP**
- Relay agent or IP helper address turns the broadcast into a unicast so one server can serve many subnets
- Scope exhaustion or unreachable server = client self-assigns 169.254.x.x (**APIPA**). Rogue **DHCP** = **DHCP** snooping

DNS RECORD	MAPS
A	Name to IPv4 address
AAAA	Name to IPv6 address
CNAME	Alias: one name to another name
MX	Mail exchanger for the domain, with priority
NS	Authoritative name servers for a zone
SOA	Start of authority: zone serial and timers
PTR	IP address back to a name (reverse zone, in-addr.arpa)
TXT	Free text: SPF , DKIM , DMARC , domain verification
SRV	Locates a service by name, port, and priority (SIP , Active Directory)

- Lookup path: recursive resolver, root, top-level domain, authoritative. Cached answers are non-authoritative and expire at the TTL
- Forward zone: names to addresses. Reverse zone: addresses to names. Primary holds the writable copy; secondaries pull by zone transfer on TCP 53
- DNSSEC signs records: integrity, not privacy. DoT (TCP 853) and DoH (443) encrypt queries: privacy
- The hosts file is checked before DNS
- NTP UDP 123, millisecond accuracy. Stratum 0 = reference clock (GPS, atomic); stratum 1 connects to it; stratum 2 syncs from 1
- PTP (IEEE 1588) = sub-microsecond with hardware timestamps. NTS = authenticated, encrypted NTP
- Clock skew over 5 minutes breaks Kerberos; wrong time breaks certificates, MFA codes, and log correlation

11 Appliances, cloud, and modern networking

DEVICE OR FUNCTION	LAYER	ONE-LINE JOB
Router	3	Connects networks, chooses paths, separates broadcast domains
Switch	2	Forwards frames by MAC; one collision domain per port
Layer 3 switch	2 and 3	Switch that routes between VLANs in hardware
Access point, WLC	2	Bridges Wi-Fi to the LAN; the controller manages many APs
Firewall	3 to 7	Stateful tracks connections; next-generation inspects applications and users
IDS		Watches a copy of traffic, alerts only. Passive, out of band
IPS		Inline in the path, blocks in real time. Can bottleneck or false-positive
VPN concentrator		Terminates many encrypted tunnels
Forward proxy	7	Requests on behalf of clients; filters, caches, hides client addresses
Reverse proxy	7	Sits in front of servers; terminates TLS, caches, protects
Load balancer	4 to 7	Spreads connections across servers: round robin, least connections, health checks
CDN		Caches content near users; cuts latency, absorbs spikes
NAS		File-level storage over SMB or NFS
SAN		Block-level storage over iSCSI or Fibre Channel; looks like a local disk
QoS		Classifies and prioritizes traffic so voice and video survive congestion

Cloud

- SaaS: use the application. PaaS: deploy your code, provider runs the OS and runtime. IaaS: rent VMs, you manage the OS
- Public (shared, pay as you go), private (dedicated), hybrid (mix)
- VPC: your isolated virtual network with its own subnets
- Network security group: stateful firewall per VM interface. Network security list or NACL: stateless rules per subnet
- Internet gateway: public addresses reach the internet. NAT gateway: private subnets reach out only
- NFV: routers, firewalls, load balancers as software instances
- Site-to-site VPN over the internet (cheap, variable) versus direct connect (private circuit, predictable, costly)
- Scalability grows capacity; elasticity grows and shrinks automatically; multitenancy shares hardware with logical isolation

SDN, SD-WAN, VXLAN, zero trust, IaC

- SDN separates the control plane (decisions) from the data plane (forwarding). Application plane on top, management plane for monitoring
- Northbound APIs: applications to controller. Southbound APIs: controller to devices
- SD-WAN: application aware, transport agnostic, zero-touch provisioning, central policy management
- VXLAN: Layer 2 frames inside UDP across Layer 3. 24-bit VNI = 16 million segments versus 4094 VLANs. Data center interconnect
- Zero trust: never trust, always verify. Every request authenticated and authorized by policy with least privilege

- **SASE** = **SD-WAN** plus cloud security (firewall, secure web gateway, **ZTNA**, **CASB**). **SSE** = the security half only
- **IaC**: playbooks and templates, source control, drift detection, dynamic inventories, automated upgrades

Physical installation

- **MDF**: main wiring room, core devices, demarcation point where the provider's responsibility ends
- **IDF**: closet per floor or wing with access switches, uplinked to the **MDF**
- 19-inch racks; 1U = 1.75 inches. Two-post for light gear, four-post or lockable cabinet for servers
- Front intake cold aisle, rear exhaust hot aisle. Match port-side intake or exhaust to the rack's airflow
- Patch panel terminates horizontal cabling; patch cords connect panel to switch. Label both ends
- **UPS** carries load through outages and conditions power; generator is long term. **PDU** is the rack power strip
- Temperature 18 to 27 C (64 to 80 F). Humidity 40 to 60 percent: too dry = static, too humid = condensation
- Clean-agent or gas fire suppression, not water. 120 V or 208/240 V; redundant supplies on separate circuits

12 Network operations

Documentation and agreements

- Physical diagram: where devices sit and cables run. Logical diagram: subnets, **VLANs**, routing, traffic flow
- Layer 1 diagram cabling; Layer 2 switches, **VLANs**, trunks, **STP**; Layer 3 routers, subnets, routing protocols
- Rack diagram: front view by rack unit. Cable map: jack to patch panel port to switch port
- Asset inventory: model, serial, location, owner, warranty, licensing. **IPAM**: every subnet and address
- **SLA**: what the provider guarantees (uptime, repair time) and penalties. **MOU**: non-binding intent. **NDA**: confidentiality
- Update documentation as the last step of every change

Life cycle, change, and configuration

- **EOL**: vendor stops selling. **EOS**: no more patches or support; replace it
- Change process in order: request, assess impact and write a rollback plan, approve, schedule a maintenance window and notify, implement and test, document
- Emergency changes take a shorter path but are still recorded. Service request = routine pre-approved work
- Production config (running now), backup config (saved before and after changes), baseline or golden config (approved standard). Drift = deviation from golden

MONITORING	DETAIL
SNMP	Agent on the device, MIB database, OID per variable. Manager polls with Get on UDP 161; device sends traps on UDP 162
SNMP versions	v1 and v2c: plaintext community string (v2c adds bulk). v3: authentication, integrity, encryption
Flow data	NetFlow, sFlow, IPFIX : who talked to whom, ports, volume. No payload. Finds top talkers
Packet capture	Protocol analyzer (Wireshark) fed by a SPAN port or an inline tap. Every byte
Syslog severity	0 Emergency, 1 Alert, 2 Critical, 3 Error, 4 Warning, 5 Notice, 6 Informational, 7 Debug
SIEM	Correlates logs from many sources, alerts on patterns, retains for compliance
Baseline	Normal utilization, latency, errors over time. Anomaly alerts fire on deviation
Solutions	Discovery (ad hoc or scheduled), traffic analysis, performance, availability (uptime versus SLA), configuration monitoring

Metrics

- Bandwidth = capacity. Throughput = what you get. Utilization = the ratio; sustained above 70 to 80 percent means congestion
- Latency = delay in **ms** (distance, queueing, satellite adds 500 to 600 **ms**). Jitter = variation in delay; choppy voice
- Packet loss = missing packets: **TCP** retransmits, voice gaps. Interface errors: **CRC**, runs, giants, discards
- Device health: **CPU**, memory, temperature, fans. **CPU** pegged often means a broadcast storm or attack

DISASTER RECOVERY	DETAIL
RPO	Recovery point objective: how much data you can lose, measured backward in time
RTO	Recovery time objective: how fast service must be restored
MTTR	Mean time to repair: average fix time
MTBF	Mean time between failures: average life before failing
Cold site	Space, power, cooling only. Days or weeks. Cheapest
Warm site	Equipment and connectivity in place; restore data. Hours to a day
Hot site	Running mirror with current data. Minutes. Most expensive
Active-active	All nodes serve; capacity drops on failure
Active-passive	Standby takes over on failure (FHRPs)
Tabletop	Walk through the plan on paper. Validation test: actually fail over or restore

Remote access and management

- Site-to-site **VPN** joins offices. Client-to-site joins remote users.
Clientless = browser over **TLS**
- Split tunnel: only corporate traffic through the **VPN**. Full tunnel: everything, inspected and protected
- Manage by **SSH** (22), **HTTPS GUI**, **API**, or console cable. Jump box or bastion: one hardened host you reach everything through
- In-band uses the production network. Out-of-band (console server, management network, cellular) works when production is down

13 Security

Concepts

- **CIA**: confidentiality (encryption, access control), integrity (hashing, signatures), availability (redundancy, **DDoS** protection)
- Vulnerability = weakness. Threat = what could exploit it. Exploit = the method. Risk = likelihood times impact
- Encryption in transit (**TLS**, **IPsec**, **SSH**) versus at rest (disk, database). **PKI**: a certificate authority binds a public key to an identity. Self-signed = browser warning
- **AAA**: authentication (who), authorization (what), accounting (record)
- **MFA** factors: something you know, have, are; plus location and time. Password plus phone app code = know plus have
- **SSO**: one login for many systems. **SAML**: web **SSO** between identity provider and applications
- **RADIUS**: **UDP** 1812/1813, network access (**Wi-Fi**, **VPN**, **802.1X**), encrypts only the password
- **TACACS+**: **TCP** 49, device administration, encrypts the whole payload, separates authentication from authorization, per-command control
- Least privilege; role-based access control; geofencing
- Honeypot = decoy system; honeynet = decoy network. Physical: locks, badges, cameras, mantraps
- Data locality laws; **PCI DSS** for card data; **GDPR** for **EU** personal data. Segment **IoT**, **IIoT**, **SCADA/ICS/OT**, guest, and **BYOD**
- **IPsec**: **AH** = integrity and authentication only; **ESP** = encryption too. Transport mode encrypts the payload; tunnel mode wraps the packet (site-to-site). **IKE** on **UDP** 500

SYMPTOM	ATTACK	DEFENSE
Service flooded from one source, or from many (botnet)	DoS, DDoS (reflection and amplification)	Upstream filtering, DDoS protection
Attacker reaches a VLAN they should not; switch spoofing or double 802.1Q tags	VLAN hopping	Disable trunk negotiation on access ports, unused native VLAN
Switch forwards all traffic to all ports; MAC table full of fakes	MAC flooding	Port security

SYMPTOM	ATTACK	DEFENSE
Forged ARP replies map the gateway IP to the attacker's MAC; traffic flows through them	ARP poisoning enabling an on-path attack	Dynamic ARP inspection
Users sent to attacker sites by forged answers or corrupted cache	DNS poisoning or spoofing	DNSSEC
Clients get a wrong gateway or DNS from an unknown server	Rogue DHCP	DHCP snooping
Forged source address	IP spoofing	Ingress filtering
Unauthorized AP on the network	Rogue access point	Wireless IDS, NAC, port security
Attacker AP broadcasting your SSID	Evil twin	Enterprise authentication, user awareness
Clients kicked off with forged management frames	Deauthentication	802.11w protected management frames
Email, phone, text, targeted, following through a door, watching a screen, trash	Phishing, vishing, smishing, spear phishing, tailgating, shoulder surfing, dumpster diving	Training, mantraps, shredding
Encrypts data for payment; hides in software; self-propagates; watches	Ransomware, trojan, worm, spyware	Backups, patching, endpoint protection
Every password tried; common words; leaked credentials reused	Brute force, dictionary, credential stuffing	MFA, lockout, unique passwords

Hardening and defense

- First steps on any device: change default passwords, disable unused ports and services, patch firmware, **SSH** and **HTTPS** only, management on its own **VLAN** or out of band
- Rotate keys and certificates; revoke compromised ones
- **NAC**: **802.1X** authentication plus posture assessment (antivirus, patches, disk encryption). Noncompliant devices go to a quarantine or remediation **VLAN**
- **ACLs**: processed top down, first match wins, implicit deny at the end
- **URL** filtering by address or category; content filtering inspects what is inside
- Firewall zones: trusted (inside), untrusted (internet), screened subnet (**DMZ**) for public-facing servers
- Segmentation with **VLANs** and firewall rules limits lateral movement

14 Troubleshooting

The seven steps, in order

1. Identify the problem: gather information, question users, identify symptoms, determine what changed, duplicate it, approach multiple problems individually.
2. Establish a theory of probable cause: question the obvious; top-to-bottom, bottom-to-top, or divide and conquer through the **OSI** layers.
3. Test the theory to determine the cause. Not confirmed: new theory or escalate.
4. Establish a plan of action to resolve the problem and identify potential effects.
5. Implement the solution or escalate as necessary.
6. Verify full system functionality and, if applicable, implement preventive measures.
7. Document findings, actions, outcomes, and lessons learned.

After the theory is confirmed, the next step is the plan, not the fix. After implementing, verify before documenting. Documenting is always last.

SYMPTOM	CAUSE
CRC errors and late collisions on one link	Duplex mismatch
Works at 80 m, fails or flaps at 120 m	Attenuation (copper limit 100 m)
Fiber link dead, strands crossed at one end	TX/RX polarity reversal
Runts (under 64 bytes)	Collisions or bad NIC
Giants (over MTU)	MTU or jumbo frame mismatch
Discards or drops rising	Congestion
Port error-disabled	Port security, BPDU guard, or flapping violation; fix cause then clear
Port administratively down	Someone typed shutdown
Camera or AP powers on then reboots, or never powers	PoE budget exceeded or wrong PoE standard
No link with SR optic on single-mode fiber	Transceiver or fiber mismatch
Interference in a factory over UTP	EMI; use STP or fiber
Broadcast storm, CPU 100 percent, MACs flapping right after a new cable	Spanning tree loop
Odd traffic paths, access switch is root	Root bridge in the wrong place; lower core priority
Host gets an address from the wrong scope after a desk move	Access port in the wrong VLAN
Untagged traffic lands in the wrong VLAN on a trunk	Native VLAN mismatch
Local devices reachable, nothing remote	Wrong or missing default gateway
Some hosts on the same subnet unreachable	Incorrect subnet mask
Intermittent connectivity, address conflict warning	Duplicate IP (static inside a scope without exclusion)
New clients get 169.254.x.x, old ones work	DHCP scope exhaustion or unreachable server
Ping by IP works, by name fails	DNS
One application or port fails, others work	ACL blocking it
Traffic goes out, replies never come back	Missing return route or asymmetric routing through a stateful firewall
Full signal bars, poor throughput	Interference or co-channel overlap; use 1, 6, 11 or move to 5 GHz
Choppy voice, delay varies	Jitter
Everything slow through one link at 100 percent	Bottleneck or congestion
Clients cling to a distant AP	Too little cell overlap or transmit power too high

COMMAND OR TOOL	SHOWS
ping	Reachability, round-trip time, loss. Loopback, self, gateway, remote in that order
tracert / traceroute	Every hop and its latency; where the path breaks. mtr and pathping repeat over time
ipconfig /all, /release, /renew, /flushdns	Windows address, mask, gateway, DNS, lease, MAC; refresh lease; clear resolver cache
ifconfig, ip addr	Linux and macOS addressing
arp -a	IP-to-MAC cache; duplicates and poisoning
nslookup, dig	Query DNS directly, pick a server, ask for a record type
netstat	Open connections and listening ports; netstat -r shows the routing table
nmap	Open ports and services; live hosts on a subnet (authorized use only)
tcpdump, Wireshark	Command-line capture; graphical protocol analyzer

COMMAND OR TOOL	SHOWS
LLDP / CDP	What device and port is on the other end of a cable
show mac-address-table	Which MAC is on which port and VLAN
show arp	The device's ARP cache
show interface	Status, speed, duplex, errors, CRCs , drops
show ip route	Routing table and how each route was learned
show vlan	VLANs and port assignments
show running-config	Current configuration
show power	PoE budget and per-port power
HARDWARE TOOL	USE
Toner and probe	Find the far end of a cable in a bundle or patch panel
Cable tester	Continuity, pinout, opens, shorts, split pairs, length
Cable certifier	Proves a run meets its category with a report
Punch-down tool, crimper	Terminate on jacks and panels; attach RJ45 plugs
Loopback plug	Tests a port's own transmit and receive
Visual fault locator	Red light shows fiber breaks and bends
OTDR	Locates fiber breaks and losses by distance
Optical power meter	Received light level
Fiber cleaning kit	Dirty connectors are the most common fiber fault
Network tap	Passive inline copy of traffic when there is no SPAN port
Wi-Fi analyzer	APs , channels, signal, dead spots
Spectrum analyzer	All radio energy including non- Wi-Fi interference
Multimeter	Voltage and continuity
PoE injector	Adds power where the switch cannot

15 Acronyms

Every acronym used in this course, 316 in all. In lessons and on this sheet, hover over an acronym for the short version; click it, or click one below, for the longer explanation.

ACRONYM	STANDS FOR	MEANING
1000BASE-LX	Gigabit Ethernet, long wavelength	1 Gbps over single-mode fiber to 10 km (also works on multimode for short runs).
1000BASE-SX	Gigabit Ethernet, short wavelength	1 Gbps over multimode fiber to about 550 m.
1000BASE-T	Gigabit Ethernet over twisted pair	1 Gbps over Cat 5e or better to 100 m, using all four pairs.
100BASE-TX	Fast Ethernet over twisted pair	100 Mbps over Cat 5 or better using two pairs.
10BASE-T	Ethernet over twisted pair	The original 10 Mbps twisted-pair Ethernet.
10GBASE-ER	10 Gigabit Ethernet, extended reach	10 Gbps over single-mode fiber to 40 km.
10GBASE-LR	10 Gigabit Ethernet, long reach	10 Gbps over single-mode fiber to 10 km.
10GBASE-SR	10 Gigabit Ethernet, short reach	10 Gbps over multimode fiber, about 300 m on OM3, 400 m on OM4.
10GBASE-T	10 Gigabit Ethernet over twisted pair	10 Gbps over Cat 6A to 100 m (Cat 6 to 55 m).
40GBASE-T	40 Gigabit Ethernet over twisted pair	40 Gbps over Cat 8 to 30 m.
4G	Fourth-generation cellular	The LTE generation of mobile broadband.

ACRONYM	STANDS FOR	MEANING
5G	Fifth-generation cellular	The current mobile network generation with higher speed and lower latency than LTE.
802.11	IEEE 802.11	The family of Wi-Fi standards.
802.1AX	IEEE 802.1AX	The current standard for link aggregation, using LACP.
802.1D	IEEE 802.1D	The original spanning tree standard.
802.1Q	IEEE 802.1Q	The VLAN tagging standard: a 4-byte tag in the Ethernet frame carries the VLAN ID on trunks.
802.1s	IEEE 802.1s	Multiple Spanning Tree Protocol.
802.1w	IEEE 802.1w	Rapid Spanning Tree Protocol.
802.1X	IEEE 802.1X	Port-based network access control: a device must authenticate before the switch port or Wi-Fi passes traffic.
802.3ad	IEEE 802.3ad	The original link aggregation standard, now part of 802.1AX.
802.3af	IEEE 802.3af	The original Power over Ethernet standard: about 15.4 W at the switch, 12.95 W at the device.
802.3at	IEEE 802.3at	PoE+: about 30 W at the switch, 25.5 W at the device.
802.3bt	IEEE 802.3bt	PoE++: 60 W (Type 3) or up to 90 W (Type 4) using all four pairs.
8P8C	Eight position, eight contact	The formal name of the connector everyone calls RJ45.
AAA	Authentication, authorization, and accounting	Prove who you are, decide what you may do, record what you did.
AAAA	Quad-A record	The DNS record that maps a name to an IPv6 address.
ACK	Acknowledgment	The TCP flag or message confirming data arrived.
ACL	Access control list	An ordered list of permit and deny rules on a router, firewall, or file system.
AD	Administrative distance	A router's trust ranking for route sources; the lowest wins when two protocols offer the same prefix.
AES	Advanced Encryption Standard	The standard symmetric cipher, with 128, 192, or 256-bit keys.
AH	Authentication Header	The IPsec protocol that provides integrity and authentication without encryption.
AP	Access point	The radio that bridges Wi-Fi clients onto the wired network.
APC	Angled physical contact	A fiber connector polish with an 8-degree angle (green) that reflects less light.
API	Application programming interface	The defined way one program talks to another, such as the web endpoints a cloud service exposes.
APIPA	Automatic Private IP Addressing	The 169.254.x.x address a host gives itself when it cannot reach a DHCP server.
ARP	Address Resolution Protocol	Finds the MAC address that goes with a known IPv4 address on the local segment.
AS	Autonomous system	A network under one administration with its own routing policy, identified by a number for BGP.
ASCII	American Standard Code for Information Interchange	The character encoding that maps letters and symbols to numbers 0 to 127.
AUP	Acceptable use policy	The rules for how staff may use company systems and data.
BASE	Baseband	The middle of Ethernet names: the cable carries one digital signal at a time.
BGP	Border Gateway Protocol	The path-vector routing protocol of the internet, between autonomous systems.
BNC	Bayonet Neill-Concelman connector	The twist-lock coax connector used on older LANs, test equipment, and CCTV.
BPDU	Bridge protocol data unit	The frames switches exchange to build the spanning tree.
BSSID	Basic service set identifier	The MAC address of one access point radio, which identifies that specific cell.
BYOD	Bring your own device	Staff use personal phones and laptops for work.
CA	Certificate authority	The trusted entity that signs certificates.
CAM	Content-addressable memory (table)	The switch's MAC address table.
CAN	Campus area network	A network spanning several buildings on one campus.
CASB	Cloud access security broker	Sits between users and cloud apps to enforce policy, visibility, and data protection.
CCMP	Counter Mode with Cipher Block Chaining Message Authentication Code Protocol	The AES-based encryption WPA2 uses.
CDN	Content delivery network	Caches content on servers around the world so users fetch it from nearby.
CDP	Cisco Discovery Protocol	Cisco's Layer 2 protocol for learning what device is on the other end of a link.
CIA	Confidentiality, integrity, and availability	The three goals of security: keep data secret, keep it correct, keep it reachable.
CIDR	Classless inter-domain routing	Prefix notation for subnets, such as /24, replacing the old A, B, C classes.
CNAME	Canonical name record	A DNS alias: one name points to another name.
CPU	Central processing unit	The processor that executes instructions.
CRC	Cyclic redundancy check	The checksum in an Ethernet frame trailer; CRC errors mean the frame was damaged in transit.
CRM	Customer relationship management	Software for tracking customers and sales, usually SaaS.
CSMA/CD	Carrier sense multiple access with collision detection	The original Ethernet rule: listen before sending, and back off if a collision is heard.
DaaS	Desktop as a service	VDI hosted by a cloud provider.

ACRONYM	STANDS FOR	MEANING
DAC	Direct attach cable	A twinaxial copper cable with transceivers permanently attached at both ends, for short 10 to 100 Gbps links.
DDoS	Distributed denial of service	Many sources, usually a botnet, flood a target at once.
DFS	Dynamic frequency selection	Wi-Fi channels in the 5 GHz band that must yield to radar.
DHCP	Dynamic Host Configuration Protocol	Hands out IP addresses, masks, gateways, and DNS servers automatically (UDP 67 and 68).
DKIM	DomainKeys Identified Mail	Outgoing mail is digitally signed; the public key is published in DNS.
DMARC	Domain-based Message Authentication, Reporting, and Conformance	A DNS policy telling receivers what to do with mail that fails SPF or DKIM, and where to send reports.
DMZ	Demilitarized zone	A screened subnet between the internet and the internal network for public-facing servers.
DNS	Domain Name System	Translates names such as www.example.com into IP addresses (UDP 53, TCP 53 for zone transfers).
DNS64	DNS 6 to 4	Synthesizes AAAA records for IPv4-only names so IPv6-only clients can reach them through NAT64.
DNSSEC	DNS Security Extensions	Digital signatures on DNS records so a resolver can tell an answer was not forged.
DoH	DNS over HTTPS	DNS queries carried inside HTTPS on TCP 443 so they cannot be read or altered in transit.
DORA	Discover, Offer, Request, Acknowledge	The four-step DHCP lease exchange.
DoS	Denial of service	One source floods or crashes a target so it cannot serve users.
DoT	DNS over TLS	DNS queries encrypted with TLS on TCP 853.
DR	Disaster recovery	The plans and sites for restoring IT after a major outage.
DSS	Data Security Standard	The PCI DSS rules for protecting payment card data.
EGP	Exterior gateway protocol	A routing protocol between organizations; today that means BGP.
EIGRP	Enhanced Interior Gateway Routing Protocol	Cisco's advanced distance-vector protocol with fast convergence and a composite metric.
EMI	Electromagnetic interference	Noise from motors, power lines, fluorescent lights, and radios that corrupts signals on copper cable.
EOL	End of life	The vendor stops selling the product.
EoS	End of support	The vendor stops providing patches and help.
ER	Extended reach	The single-mode 10 Gbps optic for 40 km.
ESP	Encapsulating Security Payload	The IPsec protocol that encrypts and authenticates the payload.
ESSID	Extended service set identifier	The SSID shared by several access points so clients can roam between them.
EU	European Union	The political union whose GDPR privacy law reaches any company holding its residents' data.
EUI-64	Extended unique identifier, 64-bit	A way to build an IPv6 interface ID from a 48-bit MAC address.
FCC	Federal Communications Commission	The US regulator of radio spectrum, including Wi-Fi bands and transmit power.
FHRP	First-hop redundancy protocol	Lets two routers share one virtual gateway address so hosts keep working if one fails.
FIN	Finish flag	The TCP flag that closes a connection gracefully.
FTP	File Transfer Protocol	Plaintext file transfer on TCP 21 (control) and 20 (data); replace with SFTP or FTPS.
FTPS	FTP Secure	FTP wrapped in TLS, ports 990 and 989 (implicit) or 21 with an upgrade (explicit).
GBIC	Gigabit interface converter	The older, larger hot-swappable transceiver replaced by the SFP.
Gbps	Gigabits per second	Billions of bits per second; gigabit Ethernet is 1 Gbps.
GDPR	General Data Protection Regulation	The European Union's privacy law for personal data of EU residents.
GG45	GG45 connector	A backward-compatible Cat 7 and Cat 8 connector that also accepts RJ45 plugs.
GHz	Gigahertz	Billions of cycles per second; CPU clock speed and Wi-Fi band frequency.
GLBP	Gateway Load Balancing Protocol	Cisco's first-hop redundancy that also spreads traffic across several gateways.
GPS	Global Positioning System	Satellite positioning, also used as a stratum 0 time reference.
GRE	Generic Routing Encapsulation	A simple tunnel that wraps one packet inside another, with no encryption.
GUI	Graphical user interface	Windows, icons, and a mouse, as opposed to a command line.
HIPAA	Health Insurance Portability and Accountability Act	The US law protecting health information and setting privacy and security rules.
HSRP	Hot Standby Router Protocol	Cisco's first-hop redundancy protocol: one active router, one standby, one virtual IP.
HTTP	Hypertext Transfer Protocol	Web traffic in the clear on TCP 80.
HTTPS	Hypertext Transfer Protocol Secure	HTTP inside TLS on TCP 443.
IaaS	Infrastructure as a service	The provider supplies virtual machines, storage, and networks; you manage the operating system and up.
IaC	Infrastructure as code	Defining networks and servers in files that tools build automatically and repeatably.
IBSS	Independent basic service set	An ad hoc Wi-Fi network between devices with no access point.
ICMP	Internet Control Message Protocol	The messaging protocol behind ping and traceroute, and behind unreachable and time-exceeded errors.

ACRONYM	STANDS FOR	MEANING
ICS	Industrial control system	The general term for computers that run physical processes; SCADA is one kind.
ID	Identifier or identification	A name or number that uniquely labels a user, device, session, or record.
IDF	Intermediate distribution frame	A smaller wiring closet on a floor or wing that feeds back to the MDF.
IDS	Intrusion detection system	Watches a copy of the traffic and alerts on attacks; cannot block them.
IEEE	Institute of Electrical and Electronics Engineers	The engineering body behind Ethernet (802.3), Wi-Fi (802.11), and many device standards.
IGMP	Internet Group Management Protocol	How hosts join and leave IPv4 multicast groups.
IGP	Interior gateway protocol	A routing protocol used inside one organization: OSPF, EIGRP, RIP, IS-IS.
IIoT	Industrial Internet of Things	Networked sensors and controllers in factories and utilities.
IKE	Internet Key Exchange	Negotiates the keys and security associations for IPsec; UDP 500, and 4500 through NAT.
IMAP	Internet Message Access Protocol	Reads mail that stays on the server, synced across devices; TCP 143, secure 993.
IMAPS	IMAP Secure	IMAP over TLS on TCP 993.
IoT	Internet of Things	Networked devices such as cameras, sensors, thermostats, and smart TVs.
IP	Internet Protocol	The Layer 3 protocol that addresses and routes packets between networks.
IPAM	IP address management	Software that tracks which addresses and subnets are in use.
IPFIX	IP Flow Information Export	The standard flow-export protocol, the successor of NetFlow.
IPS	Intrusion prevention system	Sits inline and blocks attacks as they pass.
IPsec	Internet Protocol Security	The suite that encrypts and authenticates IP packets for VPNs.
IPv4	Internet Protocol version 4	32-bit addresses written as four decimal octets, such as 192.168.1.10.
IPv6	Internet Protocol version 6	128-bit addresses in eight hex groups; no broadcast, built-in autoconfiguration.
IS-IS	Intermediate System to Intermediate System	A link-state interior routing protocol used mainly by large providers.
ISATAP	Intra-Site Automatic Tunnel Addressing Protocol	A tunneling method that carries IPv6 inside IPv4 within one organization.
iSCSI	Internet SCSI	SCSI storage commands over a TCP/IP network, giving block-level storage without Fibre Channel.
ISP	Internet service provider	The company that connects you to the internet.
IT	Information technology	Computers, networks, and the people who run them.
JPEG	Joint Photographic Experts Group (image format)	The common compressed photo format.
LACP	Link Aggregation Control Protocol	Negotiates bundling several physical links into one logical link (802.1AX, formerly 802.3ad).
LAN	Local area network	One site's network: a building, floor, or campus that one organization owns and runs.
LC	Lucent connector	The small square snap-in fiber connector, the most common on modern transceivers.
LDAP	Lightweight Directory Access Protocol	Queries a directory such as Active Directory for users and groups; TCP 389, secure 636.
LDAPS	LDAP Secure	LDAP over TLS on TCP 636.
LED	Light-emitting diode	A small light source: the status lights on a port, the lamp in cheap fiber transmitters, and modern display backlights.
LLC	Logical link control	The upper sublayer of the data link layer that identifies which network protocol a frame carries.
LLDP	Link Layer Discovery Protocol	The vendor-neutral neighbor discovery protocol (IEEE 802.1AB).
LR	Long reach	The single-mode 10 Gbps optic, 10 km.
LTE	Long-Term Evolution	4G cellular data; the workhorse mobile broadband standard.
MAC	Media access control (address)	The 48-bit hardware address burned into a network interface, used at Layer 2.
macOS	Mac operating system	Apple's desktop operating system.
MAN	Metropolitan area network	A city-sized network: bigger than a LAN, smaller than a WAN.
Mbps	Megabits per second	Millions of bits per second; the unit for network link speed.
MDF	Main distribution frame	The building's main wiring closet where outside lines and the core equipment land.
MDIX	Medium-dependent interface crossover	Auto-MDIX ports sense the cable and swap transmit and receive pairs, so crossover cables are rarely needed.
MFA	Multi-factor authentication	Two or more different kinds of proof: something you know, have, are, or where you are.
MHz	Megahertz	Millions of cycles per second; memory speed, radio channels, and older CPU clocks.
MIB	Management information base	The database of variables an SNMP agent exposes.
MIMO	Multiple input, multiple output	Using several antennas at once to send parallel data streams for more speed.
MMF	Multimode fiber	A wider core (50 or 62.5 microns) carrying many light paths, for hundreds of meters.
MOU	Memorandum of understanding	A non-binding statement of intent and roles between parties.
MPEG	Moving Picture Experts Group (video format)	The family of compressed video formats.
MPLS	Multiprotocol Label Switching	A provider WAN that forwards by short labels instead of IP lookups, with traffic engineering and QoS.
MPO	Multi-fiber push-on connector	A ribbon connector carrying 12 or 24 fibers, used for 40 and 100 Gbps parallel optics.

ACRONYM	STANDS FOR	MEANING
ms	Millisecond	One thousandth of a second; latency and response times.
MSTP	Multiple Spanning Tree Protocol	Runs one spanning tree instance for a group of VLANs instead of one per VLAN.
MTBF	Mean time between failures	The average running time before a component fails; a reliability measure.
MTP	Multi-fiber termination push-on connector	A brand-name, higher-precision version of the MPO connector.
MTTR	Mean time to repair	The average time to fix a failure and restore service.
MTU	Maximum transmission unit	The largest packet a link can carry, 1500 bytes on standard Ethernet.
MU-MIMO	Multi-user MIMO	The access point serves several clients at the same time with separate streams.
MX	Mail exchanger record	The DNS record that says which server accepts email for a domain.
NAC	Network access control	Checks a device's identity and health before letting it onto the network.
NACL	Network access control list	A stateless rule set applied to a cloud subnet.
NAS	Network-attached storage	A file server appliance that shares folders over the regular network using SMB or NFS.
NAT	Network address translation	Rewrites private inside addresses to a public address at the router so many hosts can share it.
NAT-T	NAT traversal	Wraps IPsec in UDP 4500 so it can pass through address translation.
NAT64	Network address translation 6 to 4	Lets IPv6-only hosts reach IPv4-only servers by translating between the two.
NDA	Non-disclosure agreement	A contract that keeps shared information confidential.
NDP	Neighbor Discovery Protocol	IPv6's replacement for ARP: finds neighbors' MAC addresses and routers using ICMPv6.
NFS	Network File System	The Unix and Linux file-sharing protocol.
NFV	Network functions virtualization	Running routers, firewalls, and load balancers as virtual machines instead of dedicated hardware.
NIC	Network interface card	The adapter that connects a computer to the network, wired or wireless.
NS	Name server record	The DNS record listing the authoritative name servers for a zone.
NTP	Network Time Protocol	Synchronizes clocks across the network (UDP 123); wrong time breaks Kerberos, certificates, and logs.
NTS	Network Time Security	Authentication and integrity for NTP so time cannot be spoofed.
OFDM	Orthogonal frequency-division multiplexing	The modulation that sends data across many closely spaced sub-carriers at once.
OFDMA	Orthogonal frequency-division multiple access	Wi-Fi 6 splits a channel into small sub-channels so many clients share it efficiently.
OID	Object identifier	The dotted number that names one value in an SNMP MIB.
OM1	Optical multimode 1	62.5 micron multimode fiber, orange, the oldest grade.
OM2	Optical multimode 2	50 micron multimode fiber, orange, 1 Gbps to about 550 m.
OM3	Optical multimode 3	Laser-optimized 50 micron multimode fiber, aqua, 10 Gbps to about 300 m.
OM4	Optical multimode 4	Higher-bandwidth 50 micron multimode fiber, aqua, 10 Gbps to about 400 m.
OS	Operating system	The software that manages hardware and runs applications: Windows, macOS, Linux, iOS, Android.
QSFP	Octal small form-factor pluggable	An eight-lane 400 Gbps module, slightly larger than QSFP-DD.
OSI	Open Systems Interconnection model	The seven-layer reference model used to describe where a protocol or device does its work.
OSPF	Open Shortest Path First	A link-state interior routing protocol that uses cost (bandwidth) as its metric and converges fast.
OT	Operational technology	Hardware and software that controls physical equipment, as opposed to IT.
OTDR	Optical time-domain reflectometer	A fiber tester that sends light pulses and measures reflections to find breaks and their distance.
OUI	Organizationally unique identifier	The first 24 bits of a MAC address, which identify the manufacturer.
PaaS	Platform as a service	The provider runs the platform (OS, runtime, database); you deploy code.
PAN	Personal area network	A network within arm's reach, such as Bluetooth between a phone and earbuds.
PAT	Port address translation	Many inside hosts share one public IP address, told apart by port numbers.
PC	Personal computer	A desktop or laptop for one user.
PCI	Peripheral Component Interconnect	The older expansion bus; PCIe (PCI Express) is its modern serial successor.
PDU	Protocol data unit	The name of the chunk of data at each layer: bits, frames, packets, segments or datagrams, data.
PIN	Personal identification number	A short numeric secret, a something-you-know factor.
PKI	Public key infrastructure	The system of certificate authorities, certificates, and revocation that makes public keys trustworthy.
PoE	Power over Ethernet	Delivers DC power to devices over the data cable: 802.3af 15.4 W, 802.3at 30 W, 802.3bt 60 to 90 W.
PON	Passive optical network	Fiber to the premises using unpowered splitters between the provider and many customers.
POP3	Post Office Protocol version 3	Downloads mail to one device and usually deletes it from the server; TCP 110, secure 995.
POP3S	POP3 Secure	POP3 over TLS on TCP 995.
PSK	Pre-shared key	One shared passphrase for everyone on a Wi-Fi network (Personal mode).

ACRONYM	STANDS FOR	MEANING
PTP	Precision Time Protocol	IEEE 1588 clock sync with sub-microsecond accuracy, for finance, telecom, and industrial systems.
PTR	Pointer record	The reverse DNS record: IP address back to a name.
PVC	Polyvinyl chloride	The ordinary cable jacket, which gives off toxic smoke; plenum cable is required in air spaces.
QSFP	Quad small form-factor pluggable	A four-lane module for 40 Gbps.
QSFP-DD	Quad small form-factor pluggable double density	Eight lanes for 400 Gbps.
QSFP+	Quad small form-factor pluggable plus	Four 10 Gbps lanes: a 40 Gbps module.
QSFP28	Quad small form-factor pluggable 28	Four 25 Gbps lanes: a 100 Gbps module.
RA	Router advertisement	The IPv6 message a router sends with the prefix and gateway information hosts use for SLAAC.
RADIUS	Remote Authentication Dial-In User Service	The AAA server behind Wi-Fi Enterprise, VPNs, and 802.1X; UDP 1812 and 1813.
RARP	Reverse Address Resolution Protocol	An obsolete protocol for finding an IP address from a MAC address.
RBAC	Role-based access control	Permissions are assigned to roles, and users get roles.
RC4	Rivest Cipher 4	A fast stream cipher used in WEP and old TLS; broken and deprecated.
RDP	Remote Desktop Protocol	Microsoft's remote graphical desktop on TCP 3389.
RF	Radio frequency	Radio signals; the medium of Wi-Fi, cellular, and Bluetooth, and a source of interference and, in medicine, surgical energy.
RFC	Request for Comments	The numbered documents that define internet standards, such as RFC 1918 private addresses.
RFI	Radio frequency interference	Electromagnetic interference in the radio bands, from transmitters, microwaves, and other radios.
RG-59	Radio Guide 59 coaxial cable	Thinner coax for short runs such as CCTV, often with a BNC connector.
RG-6	Radio Guide 6 coaxial cable	The modern coax grade for cable TV and cable modems, with an F-type connector.
RIP	Routing Information Protocol	The classic distance-vector protocol: hop count metric, 15-hop limit, slow.
RJ11	Registered jack 11	The small telephone plug, used for analog phone lines and DSL.
RJ45	Registered jack 45	The eight-pin plug on Ethernet twisted-pair cables (properly an 8P8C connector).
RPC	Remote procedure call	A program calls a function on another computer as if it were local.
RP0	Recovery point objective	The maximum acceptable data loss, measured in time.
RST	Reset flag	The TCP flag that aborts a connection immediately.
RSTP	Rapid Spanning Tree Protocol	The faster version of spanning tree (802.1w) that reconverges in seconds.
RT0	Recovery time objective	The maximum acceptable time to restore a service after failure.
RX	Receive	The receiving side of a link or the receiving fiber strand.
SaaS	Software as a service	Complete applications delivered over the web, such as email and office suites.
SAE	Simultaneous authentication of equals	The WPA3 Personal handshake that resists offline password guessing.
SAML	Security Assertion Markup Language	An XML standard for passing authentication between an identity provider and a web application (enterprise SSO).
SAN	Storage area network	A dedicated network that gives servers block-level access to shared disk arrays.
SASE	Secure access service edge	SD-WAN networking plus cloud-delivered security (firewall, web gateway, ZTNA, CASB) as one service.
SC	Subscriber connector	The larger square push-pull fiber connector, common on older gear and patch panels.
SCADA	Supervisory control and data acquisition	Systems that monitor and control industrial processes such as power, water, and manufacturing.
SCP	Secure Copy Protocol	Simple file copying over SSH, TCP 22.
SD-WAN	Software-defined wide area network	Steers traffic across several cheap links (broadband, LTE, MPLS) by policy from a central controller.
SDN	Software-defined networking	Separates the control plane (decisions) from the data plane (forwarding) and centralizes control.
SFP	Small form-factor pluggable transceiver	The hot-swappable module that gives a switch port its fiber or copper interface, 1 Gbps.
SFP+	Enhanced small form-factor pluggable	The 10 Gbps version of the SFP module.
SFTP	SSH File Transfer Protocol	File transfer inside an SSH session, TCP 22.
SHA-256	Secure Hash Algorithm 256-bit	The standard hash today: a 256-bit fingerprint of any input.
SIEM	Security information and event management	Collects logs from everything, correlates them, and raises alerts.
SIP	Session Initiation Protocol	Sets up and tears down voice and video calls; port 5060, or 5061 with TLS.
SLA	Service level agreement	The contract stating measurable service commitments, such as 99.9% uptime.
SLAAC	Stateless address autoconfiguration	IPv6 hosts build their own address from the router's advertised prefix, no DHCP server needed.
SMB	Server Message Block	Windows file and printer sharing on TCP 445.
SMF	Single-mode fiber	A tiny core (about 9 microns) and a laser, for long distances.

ACRONYM	STANDS FOR	MEANING
SMTP	Simple Mail Transfer Protocol	Sends email between servers on TCP 25; clients submit on 587 with TLS.
SMTPS	SMTP Secure	SMTP with TLS, port 465 (implicit) or 587 with STARTTLS.
SNMP	Simple Network Management Protocol	Polls and receives alerts from network devices (UDP 161 polls, 162 traps).
SNTP	Simple Network Time Protocol	A lighter version of NTP for clients that need less accuracy.
SOA	Start of authority record	The record at the top of a DNS zone: primary server, contact, serial number, and timers.
SOX	Sarbanes-Oxley Act	The US law requiring accurate financial reporting and controls, with IT audit implications.
SPAN	Switched port analyzer	A switch feature that mirrors traffic from ports or VLANs to a monitoring port.
SPF	Sender Policy Framework	A DNS TXT record listing which servers may send mail for a domain.
SQL	Structured Query Language	The language used to query databases; also the name of the servers on TCP 1433 (Microsoft) and 3306 (MySQL).
SR	Short reach	The multimode 10 Gbps optic, about 300 m on OM3.
SRV	Service record	A DNS record that locates a service: host, port, priority, and weight.
SSE	Security service edge	The security half of SASE: secure web gateway, CASB, zero trust access, delivered from the cloud.
SSH	Secure Shell	Encrypted remote command-line access and file transfer on TCP 22; replaces Telnet and FTP.
SSID	Service set identifier	The Wi-Fi network name that access points broadcast and clients join.
SSO	Single sign-on	Log in once and reach many systems without signing in again.
ST	Straight tip connector	The round bayonet twist-lock fiber connector; legacy.
STP	Spanning Tree Protocol	Blocks redundant switch links so frames cannot loop, while keeping them ready as backups.
SVI	Switch virtual interface	A virtual Layer 3 interface on a switch that gives a VLAN an IP address and routes for it.
SYN	Synchronize flag	The first packet of a TCP handshake.
SysLog	System logging protocol	Sends device log messages to a central server on UDP 514 with severity levels 0 to 7.
T568A	TIA/EIA 568A wiring standard	One of the two pin-outs for RJ45 cables; green pair on pins 1 and 2.
T568B	TIA/EIA 568B wiring standard	The more common RJ45 pin-out; orange pair on pins 1 and 2.
TACACS+	Terminal Access Controller Access-Control System Plus	Cisco's AAA protocol for device administration on TCP 49; encrypts the whole payload and separates authentication from authorization.
TCP	Transmission Control Protocol	Reliable, ordered, connection-based transport: three-way handshake, acknowledgments, retransmission.
TERA	TERA connector	A shielded, non-RJ45 connector used with Cat 7 cabling.
TFTP	Trivial File Transfer Protocol	Bare-bones UDP 69 file transfer with no login, used for firmware and config files.
TKIP	Temporal Key Integrity Protocol	WPA's per-packet keying scheme; deprecated.
TLS	Transport Layer Security	The encryption layer under HTTPS and most secure protocols; use 1.2 or 1.3.
TOTP	Time-based one-time password	A six-digit code from an authenticator app that changes every 30 seconds.
TTL	Time to live	A counter that keeps packets and DNS records from living forever.
TV	Television	A display or broadcast receiver; smart TVs are IoT devices.
TX	Transmit	The sending side of a link or the sending fiber strand.
TXT	Text record	A free-text DNS record, used for SPF, DKIM, DMARC, and domain ownership checks.
UDP	User Datagram Protocol	Connectionless, fast transport with no delivery guarantee; used for voice, video, DNS, DHCP.
UPC	Ultra physical contact	The flat, blue fiber connector polish used in most LANs.
UPS	Uninterruptible power supply	A battery unit that keeps equipment running through a power cut and conditions the power.
URL	Uniform resource locator	A web address such as https://www.example.com/page .
US	United States	The country; US standards, 60 Hz mains, and US connector conventions.
USB	Universal Serial Bus	The standard port for peripherals, storage, and charging.
UTP	Unshielded twisted pair	The standard copper network cable: four twisted pairs, no shield.
VA	Volt-ampere	The unit of apparent power used to size a UPS.
VCSEL	Vertical-cavity surface-emitting laser	The cheap laser used in multimode fiber transceivers.
VFL	Visual fault locator	A visible red laser that lights up a fiber so you can see breaks and trace strands.
VLAN	Virtual local area network	A logical LAN on a switch: ports in different VLANs are separate broadcast domains.
VLSM	Variable-length subnet masking	Using different subnet sizes inside one network so addresses are not wasted.
VM	Virtual machine	A complete computer running as software on a hypervisor.
VNI	VXLAN network identifier	The 24-bit segment ID in a VXLAN header.
VoIP	Voice over IP	Telephone calls carried as packets on the data network.
VPC	Virtual private cloud	Your isolated virtual network inside a cloud provider.
VPN	Virtual private network	An encrypted tunnel across an untrusted network such as the internet.

ACRONYM	STANDS FOR	MEANING
VRRP	Virtual Router Redundancy Protocol	The open-standard first-hop redundancy protocol.
VTP	VLAN Trunking Protocol	Cisco's protocol for distributing VLAN definitions to other switches.
VXLAN	Virtual extensible LAN	Tunnels Layer 2 segments over a Layer 3 network with a 24-bit ID, giving 16 million segments instead of 4094 VLANs.
WAN	Wide area network	Connects LANs across cities or countries, usually over links leased from a provider.
WEP	Wired Equivalent Privacy	The original Wi-Fi encryption, broken in minutes; never use it.
Wi-Fi	Wireless fidelity	The trade name for IEEE 802.11 wireless networking.
WLAN	Wireless local area network	A LAN delivered over Wi-Fi instead of cables.
WLC	Wireless LAN controller	A central device that configures and manages many lightweight access points.
WPA	Wi-Fi Protected Access	The interim fix for WEP that used TKIP; deprecated.
WPA2	Wi-Fi Protected Access 2	Wi-Fi security using AES with CCMP; still common and acceptable.
WPA3	Wi-Fi Protected Access 3	The current Wi-Fi security standard: SAE in Personal mode, stronger encryption, protected management frames.
ZTNA	Zero trust network access	Every request is authenticated and authorized, regardless of where it comes from.